

Last Change	Review	Approval

Document History

Version	Date	Changes	JIRA-Ticket	Name
V1	2025-09-11	Release of v1.0.1 of gammaSTAR Reconstructions.	n/a	jhu
V2	2025-09-25	Release of v1.0.2 to fix bugs for non-isotropic matrix sizes and phase correction for GRASE sequences.	n/a	jhu
V3	2025-11-04	Release of v1.0.3 which improves the code documentation, removes unnecessary functions and fixes crashes for non-imaging sequences.	n/a	jhu
V4	2026-02-09	Release of v1.1.0 which adds exported sequences as DICOM tags and changes handling of non-Cartesian data.	n/a	jhu
V5	2026-04-23	Release of v1.1.1 which removes redundant functionality and improves the code quality.	n/a	jhu

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1 Introduction and Overview

This document summarizes the delivery of **gammaSTAR Reconstructions** for generic image reconstruction from MR raw data simulated using the publicly available gammaSTAR frontend. The software handles various k-space acquisition strategies such as 2D or 3D Cartesian sequences with/without ramp sampling, 2D or 3D radial or spiral sequences as well as PROPELLER trajectories.

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2 Release Summary

Software Component	gammaSTAR Reconstructions
Version	https://gitlab.fme.lan/gammastar/gammastar_recon/-/tags/v1.1.1_release SHA: b370a5ac4bf857100ec30b2b24a62e8b7a3ce411
Embedding Product(s)	Not applicable
Purpose of this Release	Provision of the gammaSTAR Reconstructions v1.1.1 software under AGPLv3-clause license for research and educational purposes. Not for clinical use.
OS and Compiler Details	OS: Windows 11 Python Version: 3.12
Embedded Libraries	The following libraries are used by the software. Detailed license information is contained in the delivered "third_party_licenses" folder and in the NOTICE file. • MRI Nufft v1.4.0 (BSD-3-Clause License) • Python ISMRMRD Server (MIT License) • pydicom v2.4.5 (MIT-based License) • ismrmrd v1.14.2 (ISMRMRD SOFTWARE LICENSE JULY 2013) • sigpy v0.1.27 (BSD-2-Clause License) • xmltodict v0.14.2 (MIT License) • numpy v2.2.6 (NumPy License) • scikit-learn v1.7.2 (MIT License)
Functionality	The software provides a unified reconstruction solution for arbitrary MR sequences implemented using the gammaSTAR framework. Cartesian data as well as non-Cartesian data in 2D or 3D can be reconstructed as well as PROPELLER sequences. No input by the user is required if data was acquired using the gammaSTAR framework. GammaSTAR Reconstructions v1.1.1 improves v1.1.0 by the following measures: • Removed unnecessary functionality to make software more reactive and to improve the user experience. • Improved code quality. • Improved image reconstruction speed.
Regulatory Compliance	THIS SOFTWARE IS PROVIDED "AS IS" AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED

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Safety Classification	None
Risk Measures	No risk measures defined.
Known Nonconformities and Restrictions	The software is not qualified for use as a medical product or as part thereof. No bugs or restrictions are known.